

LAWTON FORT SILL NORTH, OK, USA

WMO: 723550

Lat: 34.650N

Lon: 98.400W

Elev: 1189

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	20.1	2.7	6.6	20.5	7.5	8.4	25.2	29.6	36.8	26.9	37.4	10.6	10	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	103.6	72.9	100.4	73.5	97.6	73.7	79.8	92.5	78.2	90.8	77.1	89.6	10.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	144.3	86.7	75.0	137.0	84.6	73.5	130.5	82.7	44.3	92.4	42.6	90.5	41.3	89.5	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	23.2	20.3	8.6	106.2	4.5	3.7	5.3	108.8	2.7	111.0	0.1	113.1	-3.2	115.7
			7.2	80.5	4.0	2.2	4.4	82.1	2.1	83.3	-0.2	84.6	-3.1	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	40.8	45.3	53.7	61.7	70.6	80.4	84.4	83.3	75.8	63.8	51.9	42.3
	DBStd	17.48	10.01	11.13	10.34	8.08	7.51	5.50	5.11	5.32	7.41	8.56	8.70	9.28
	HDD50	955	310	197	80	11	0	0	0	0	0	9	79	269
	HDD65	3094	749	553	369	153	33	0	0	0	11	125	397	704
	CDD50	5680	26	67	194	361	640	912	1066	1033	775	438	137	31
	CDD65	2343	0	3	18	53	208	462	601	568	336	89	5	0
	CDH74	28409	9	54	223	607	2087	5554	8123	7226	3568	882	70	6
	CDH80	15000	0	11	50	171	830	2920	4817	4157	1759	277	8	0
Wind	WSAvg	10.0	10.1	10.7	12.0	11.8	10.5	10.3	8.8	8.5	8.7	9.7	9.9	9.6
Precipitation	PrecAvg	29.1	1.1	1.3	2.2	2.5	4.4	3.8	2.0	2.5	3.3	2.8	1.8	1.5
	PrecMax	41.7	3.2	4.6	6.8	6.2	16.3	10.3	8.4	9.4	7.5	11.1	6.1	5.8
	PrecMin	16.4	0.0	0.0	0.0	0.1	0.2	0.6	0.0	0.0	0.4	0.2	0.0	0.0
	PrecStd	7.4	0.9	1.0	1.7	1.7	3.2	2.5	1.8	2.2	2.0	2.7	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	80.3	85.6	90.2	97.0	104.1	107.7	107.4	101.8	91.6	81.0	73.3
		MCWB	55.7	58.4	62.3	64.3	69.7	72.7	72.9	72.0	71.2	70.0	63.7	57.1
	2%	DB	67.8	74.3	79.9	84.7	92.3	99.4	103.9	103.1	97.2	86.5	75.4	67.3
		MCWB	52.4	56.8	62.5	64.3	71.7	73.2	72.7	74.1	72.4	68.2	61.6	55.8
	5%	DB	62.8	68.4	75.6	80.9	88.4	96.1	100.6	99.9	93.5	82.7	71.0	62.3
		MCWB	50.3	55.4	60.3	63.8	70.8	73.7	73.5	74.3	72.9	66.7	58.6	52.7
	10%	DB	57.9	63.2	71.1	77.0	84.7	93.0	97.8	96.6	89.9	79.0	66.7	57.5
		MCWB	47.4	52.0	57.9	62.7	70.2	74.8	74.1	74.0	71.9	64.7	56.7	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	65.3	68.9	72.2	78.2	81.5	81.5	81.5	79.5	75.9	67.9	64.1
		MCDB	68.9	70.5	77.7	81.1	87.2	92.5	94.3	97.6	89.2	83.4	74.0	67.7
	2%	WB	55.8	61.4	66.2	69.6	75.7	79.1	79.5	79.0	77.4	72.6	64.3	58.8
		MCDB	62.7	68.2	74.2	77.7	84.6	90.6	93.2	92.6	88.7	81.0	71.0	63.1
	5%	WB	51.5	57.8	63.3	67.8	73.7	77.4	78.0	77.3	75.5	70.0	61.8	54.1
		MCDB	60.6	64.6	71.4	75.7	83.7	89.3	91.6	90.6	86.3	77.2	68.0	59.9
	10%	WB	47.9	53.6	60.2	65.8	71.8	76.0	76.7	76.0	73.9	67.7	58.8	49.7
		MCDB	57.0	63.2	68.6	73.5	81.4	88.2	89.9	89.0	84.2	76.0	65.3	56.9
Mean Daily Temperature Range	5% DB	MDBR	23.4	23.1	23.5	23.4	21.4	21.5	22.9	22.7	22.0	23.4	23.1	20.7
		MCDBR	32.1	32.0	29.5	29.3	26.0	25.7	27.1	26.8	26.7	27.6	28.8	29.2
		MCWBR	18.9	18.1	15.6	14.6	10.5	8.2	6.8	7.1	8.4	12.7	16.9	18.0
	5% WB	MCDBR	28.5	27.1	24.6	22.9	21.8	21.6	22.8	22.5	21.4	21.5	24.2	24.5
		MCWBR	18.3	18.4	15.1	14.2	11.4	9.3	8.5	7.9	8.8	12.9	17.0	18.0
Clear-Sky Solar Irradiance	taub	0.289	0.298	0.335	0.369	0.405	0.425	0.421	0.417	0.384	0.347	0.311	0.292	
	taud	2.520	2.511	2.423	2.353	2.323	2.306	2.352	2.307	2.373	2.466	2.500	2.511	
	Ebn at Noon	294	303	298	291	280	273	273	273	277	281	283	286	
	Edh at Noon	26	29	35	39	41	42	39	41	36	30	26	25	
All-Sky Solar Radiation	RadAvg	935	1124	1491	1826	1963	2179	2192	2005	1666	1277	984	793	
	RadStd	95	160	95	146	209	189	129	116	134	176	125	87	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page